

The Renaming of the OMI Resolver Authorities

Tetragrammatron Relation Governor, Metatron Incidence Scribe, Gnomonic Projection Azimuth, Omicron, and Omnicron

Status

Canonical renaming doctrine.

This document replaces the older authority naming where relation validation, incidence tracking, balanced witnessing, projection, and runtime coordination had accumulated under names that no longer matched their actual computational roles.

The renaming is not cosmetic.

The names appear throughout the protocol, code, proof fixtures, diagrams, runtime descriptions, and conceptual model. A wrong name does more than create confusion. It changes how later components are interpreted, which authority appears to own a witness, and which layer seems permitted to validate, record, project, or carry state.

The corrected authority split is:

```
Tetragrammatron Relation Governor
  adjudicates relation closure

Metatron Incidence Scribe
  records gauge incidence and carry

Gnomonic Projection Azimuth
  orients balanced projected contrast

Omicron
  defines the bounded low gauge

Omnicron
  resolves the full low/high circular runtime
```

The central lock is:

```
Relation is not incidence.

Incidence is not projection.
```

Projection is not validation.

Gauge is not runtime.

Runtime is not acceptance.

1. Why the Renaming Is Necessary

The old structure had two different functions inside the Tetragrammatron.

The first function worked over the Polybius nibble field:

```
0 1 2 3
4 5 6 7
8 9 A B
C D E F
```

It checked:

```
diagonal XOR closure
equal diagonal sums
centered fold
shifted fold
relation carry-forward
```

Its witnesses were:

```
0x1E
0x3C
0x78
```

The second function worked over the four gauges:

```
FS
GS
RS
US
```

It checked:

```
gauge order
open/sealed polarity
pairwise incidence
hexadecimal-place stepping
gauge carry
```

Its witnesses were:

```
0x18
0x0001
0x0010
0x0100
0x1000
0x10000
```

These two functions cooperate, but they do not perform the same kind of computation.

The first answers:

```
Does this relation close?
```

The second answers:

```
Where does this relation occur,
under which polarity,
through which gauge,
and with what carry?
```

The old naming treated both as Tetragrammatron functions.

The corrected naming keeps only the closure machine under Tetragrammatron and moves the incidence machine to Metatron.

The old Metatron witness:

```
0xAA55
0x55AA
```

does not primarily describe incidence inscription.

It describes:

balanced alternation
complementary contrast
observer-facing orientation
projection-axis symmetry

That role is better named:

Gnomonic Projection Azimuth

The renaming therefore follows the actual mathematics.

2. The Five Principal Components

The corrected system has five principal components.

1. Tetragrammatron Relation Governor
2. Metatron Incidence Scribe
3. Gnomonic Projection Azimuth
4. Omicron Gauge
5. Omnicron Runtime Resolver

Their roles are:

Tetragrammatron
closes

Metatron
inscribes

Gnomonic Projection Azimuth
orients

Omicron
gauges

Omnicon
resolves

Receipt remains separate:

Receipt
records acceptance

3. Tetragrammatron Relation Governor

3.1 Definition

The Tetragrammatron Relation Governor is the finite grammar computron that adjudicates whether a declared relation closes under the Polybius nibble laws.

Its authoritative witness family is:

0x1E
0x3C
0x78

Its role is:

diagonal closure
centered fold
shifted/full-system fold
relation carry-forward

It does not own gauge incidence.

It does not own projection.

It does not own the balanced `0xAA55 / 0x55AA` witness.

It owns the closure ladder.

3.2 The Polybius Field

The nibble field is:

0 1 2 3
4 5 6 7

8	9	A	B
C	D	E	F

The two principal diagonals are:

$D^+ = \{0, 5, A, F\}$
 $D^- = \{3, 6, 9, C\}$

Their XOR closures are:

$0 \wedge 5 \wedge A \wedge F = 0x00$
 $3 \wedge 6 \wedge 9 \wedge C = 0x00$

Binary form:

D^+ :
0000
0101
1010
1111

0000

 D^- :
0011
0110
1001
1100

0000

Each diagonal contains a balanced nibble traversal.

Each diagonal begins and ends in different visible positions, but the complete XOR path cancels.

This is why the relation remains inspectable without being destroyed.

Canonical law:

A closed relation may preserve every member
while its total XOR witness returns to zero.

3.3 The 0x1E Relation Witness

The sums of the two diagonals are:

$$\text{SUM}(D^+) = 0 + 5 + 10 + 15 = 30 = 0x1E$$

$$\text{SUM}(D^-) = 3 + 6 + 9 + 12 = 30 = 0x1E$$

Thus:

0x1E

is the single-diagonal relation witness.

It is not the XOR result.

The XOR result is zero.

The sum witness preserves the finite magnitude of the closed relation.

This distinction matters:

XOR closure
proves cancellation

SUM witness
preserves relational weight

The Tetragrammatron Relation Governor adjudicates both.

3.4 The 0x3C Centered Fold

The full diagonal set is:

$D^+ \cup D^-$
=
{0, 3, 5, 6, 9, A, C, F}

Its sum is:

$0x1E + 0x1E = 0x3C$

The complement is:

$K = \{1, 2, 4, 7, 8, B, D, E\}$

and:

$SUM(K) = 0x3C$

This produces a centered equality:

$SUM(diagonals) = SUM(complement) = 0x3C$

Therefore:

$0x3C$

is the centered fold witness.

It is centered because the selected diagonal structure and its complement carry equal total weight.

The fold does not mean every element is identical.

It means the two partitions balance under the chosen invariant.

Canonical law:

The center is not sameness.

The center is equal carried weight across distinct partitions.

3.5 The 0x78 Shifted Full-System Witness

The complete nibble field sums to:

$$0 + 1 + 2 + \dots + 15 = 120 = 0x78$$

Since:

$$0x3C \ll 1 = 0x78$$

the centered fold shifts into the full-system witness.

Thus the relation ladder is:

0x1E
single relation diagonal

0x3C
paired centered fold

0x78
shifted full-system fold

The laws are:

$$\begin{aligned} 0x1E + 0x1E &= 0x3C \\ 0x3C \ll 1 &= 0x78 \\ 0x78 \gg 1 &= 0x3C \\ 0x3C \wedge 0x78 &= 0x44 \end{aligned}$$

Binary:

$$\begin{aligned} 0x3C &= 00111100 \\ 0x78 &= 01111000 \\ 0x44 &= 01000100 \end{aligned}$$

Both `0x3C` and `0x78` contain four one bits and four zero bits.

The shift changes position without changing population balance.

That is why `0x78` is a carry-forward witness rather than a new unrelated constant.

3.6 Tetragrammatron Canon

Tetragrammatron Relation Governor:

`0x1E`

diagonal relation witness

`0x3C`

centered fold witness

`0x78`

shifted/full-system fold witness

Role:

validate relation closure

validate balanced partition

validate carry-forward

The canonical sentence is:

Tetragrammatron adjudicates whether relation closes,
whether the closed partitions balance,
and whether the balanced fold carries forward lawfully.

4. Metatron Incidence Scribe

4.1 Definition

Metatron becomes the Incidence Scribe.

Its authoritative witness family is:

`0x18`

`0x0001`

```
0x0010
0x0100
0x1000
0x10000
```

Its role is:

```
gauge incidence
open/sealed polarity
hexadecimal-place stepping
gauge carry
incidence inscription
```

Metatron does not decide Polybius relation closure.

Metatron does not own `0x1E`, `0x3C`, or `0x78`.

Metatron records where a relation occurs and how it traverses the gauge structure.

4.2 The Four Gauges

The four visible gauge offsets are:

```
FS = 0x0001
GS = 0x0010
RS = 0x0100
US = 0x1000
```

These are not arbitrary labels attached to four values.

They are four hexadecimal place positions.

Binary form:

```
FS = 0000 0000 0000 0001
GS = 0000 0000 0001 0000
RS = 0000 0001 0000 0000
US = 0001 0000 0000 0000
```

The order is:

FS < GS < RS < US

The conceptual stepping notation is:

FS $\ll_4$ GS $\ll_4$ RS $\ll_4$ US $\ll_4$ CARRY

where:

$\ll_4$
=
advance four binary places
=
advance one hexadecimal place

The executable stepping is:

0x0001 $\ll$ 4 = 0x0010
0x0010 $\ll$ 4 = 0x0100
0x0100 $\ll$ 4 = 0x1000
0x1000 $\ll$ 4 = 0x10000

Therefore:

CARRY = 0x10000

This is the gauge-place carry.

It is not the same as:

0x3C $\ll$ 1 = 0x78

which is the relation fold-carry.

4.3 The 0x18 Gauge-Incidence Witness

The low Omicron character ring may be divided into four 32-position sectors:

```
0x00..0x1F
0x20..0x3F
0x40..0x5F
0x60..0x7F
```

For incidence reading, each sector divides into:

```
28-position body
+
4-position terminal hinge
```

The interval pairs are:

```
(0x00 . 0x1B)
(0x1C . 0x1F)

(0x20 . 0x3B)
(0x3C . 0x3F)

(0x40 . 0x5B)
(0x5C . 0x5F)

(0x60 . 0x7B)
(0x7C . 0x7F)
```

The first hinge is exactly:

```
0x1C FS
0x1D GS
0x1E RS
0x1F US
```

Each sector produces:

```
0x00 ^ 0x1B ^ 0x1C ^ 0x1F = 0x18

0x20 ^ 0x3B ^ 0x3C ^ 0x3F = 0x18
```

$$0x40 \wedge 0x5B \wedge 0x5C \wedge 0x5F = 0x18$$

$$0x60 \wedge 0x7B \wedge 0x7C \wedge 0x7F = 0x18$$

Thus:

$$0x18 = 24 \text{ decimal} = 4!$$

This witness survives sector rotation because the high sector prefix appears four times and cancels under XOR.

Let:

$$B \in \{0x00, 0x20, 0x40, 0x60\}$$

Then:

$$\begin{aligned} &B \\ &\wedge (B \wedge 0x1B) \\ &\wedge (B \wedge 0x1C) \\ &\wedge (B \wedge 0x1F) \\ &= \\ &0x18 \end{aligned}$$

because:

$$B \wedge B \wedge B \wedge B = 0$$

The sector changes.

The incidence pattern does not.

4.4 Open and Sealed Polarity

The four gauges are read under two states:

```
open
sealed
```

This gives eight incidence points:

```
open FS
open GS
open RS
open US

sealed FS
sealed GS
sealed RS
sealed US
```

The six pairwise gauge planes are:

```
FS-GS
FS-RS
FS-US
GS-RS
GS-US
RS-US
```

The incidence counts are:

```
points = 8
circles = 6
point degree = 3
circle degree = 4
flags = 24
```

Therefore:

```
8 × 3 = 24
6 × 4 = 24
```

The same number appears in three related forms:

```
24
0x18
4!
```

These are not three unrelated coincidences.

They are three notations for the same finite incidence count:

```
24 flags

0x18 machine witness

4! permutations of four ordered gauges
```

Metatron records this incidence structure.

4.5 Why Metatron Is the Scribe

The incidence structure does not determine relation truth by itself.

It records:

```
which gauge
which polarity
which pairwise plane
which place-value step
which carry position
```

That is a scribal operation.

It preserves the route through which a relation was encountered.

It does not replace the Tetragrammatron's closure test.

Canonical distinction:

```
Tetragrammatron asks:
Does the relation close?
```


Metatron asks:
Where and under what incidence did it occur?

Metatron therefore owns the incidence record, not the relation verdict.

4.6 Metatron Canon

Metatron Incidence Scribe:

0x18

0x0001

0x0010

0x0100

0x1000

0x10000

Role:

record FS/GS/RS/US incidence
record open/sealed polarity
record pairwise gauge planes
record hexadecimal-place stepping
record gauge carry

The canonical sentence is:

Metatron scribes the place, polarity, incidence, and carry
through which a relation becomes addressable.

5. Gnomonic Projection Azimuth

5.1 Definition

The old Metatron balanced witness becomes:

Gnomonic Projection Azimuth

Its witness family is:

```
0xAA55  
0x55AA
```

Its role is:

```
alternating balanced witness  
complementary contrast  
observer-facing orientation  
projection azimuth  
scribe-facing display
```

It does not validate relation closure.

It does not record gauge incidence.

It does not accept state.

It determines how an accepted and inscribed relation is oriented for inspection.

5.2 The Balanced Patterns

The two patterns are:

```
0xAA55 = 10101010 01010101  
  
0x55AA = 01010101 10101010
```

Each contains:

```
8 ones  
8 zeros
```

They are exact complements:

```
0xAA55 ^ 0x55AA = 0xFFFF
```

and:

```
0xAA55 ^ 0xFFFF = 0x55AA
```

```
0x55AA ^ 0xFFFF = 0xAA55
```

The patterns show maximal local alternation while preserving global balance.

Every adjacent step displays contrast.

The complete word preserves equality of high and low population.

This is why these values belong to a projection axis.

5.3 Why Gnomonic

A gnomonic projection sends rays from the center of a sphere to a tangent plane.

Its defining property is:

```
great-circle routes become straight lines
```

The projected route becomes inspectable without being confused with the original curved surface.

This matches the OMI projection boundary:

```
bounded relation  
→ observer-facing tangent  
→ readable straight route
```

The tangent plane is not the original relation.

The projected line is not the validating authority.

The azimuth only identifies the observer-relative angular orientation of the projection.

5.4 Why Azimuth

Azimuth is a local angular coordinate around an observer-facing reference axis.

In this protocol, it describes:

where the projected relation appears
relative to the projection axis

The two balanced words provide complementary orientations:

0xAA55
one alternating phase

0x55AA
the complementary alternating phase

Their XOR gives the full complement field:

0xFFFF

Therefore the Gnomonic Projection Azimuth can expose:

orientation
contrast
complement
phase reversal
observer-relative crossing

without becoming a validator or scribe.

5.5 Gnomonic Projection Azimuth Canon

Gnomonic Projection Azimuth:

0xAA55
0x55AA

Role:

orient alternating contrast
preserve balanced complement
project accepted incidence
expose observer-relative azimuth

The canonical sentence is:

The Gnomonic Projection Azimuth turns a bounded accepted relation into an observer-facing orientation without converting projection into authority.

6. Omicron

6.1 Definition

Omicron is the bounded low gauge.

It owns:

- the canonical low gauge table
0x00..0x7F
- the four 32-position character sectors
- the hidden-to-readable tangent
- the F-column branch
- the character and place-value coupling

Its four character sectors are:

- 0x00..0x1F
- 0x20..0x3F
- 0x40..0x5F
- 0x60..0x7F

Omicron provides the finite sign surface.

6.2 Hidden-to-Readable Tangent

The hidden control region ends at:

0x1F US

The readable region begins at:

```
0x20 SP
```

The F-column branch is:

```
0x1F US  
0x2F /  
0x3F ?  
0x4F 0  
0x5F _  
0x6F o
```

This produces:

```
US/  
→ ?0_o
```

and unfolds:

```
omi---imo
```

into:

```
omi---imo?0_o
```

The branch is a traversal through the gauge, not a parser permission flag.

6.3 Place-Value Coupling

The outer hexadecimal lanes are:

```
0x0000  
0x1000  
0x2000  
0x3000  
  
0x4000  
0x5000
```

```
0x6000
0x7000

0x8000
0x9000
0xA000
0xB000

0xC000
0xD000
0xE000
0xF000
```

The inner Omicron ring chooses the sign.

The outer lane chooses its place.

The combined position is:

```
position =
high-nibble lane
OR
low character
```

Conceptually:

```
sign × place
```

Omicron therefore defines the bounded sign-and-place gauge.

6.4 Omicron Canon

```
Omicron:

0x00..0x7F

Role:

define the bounded character gauge
define the hidden/readable tangent
```

```
define the F-column branch  
couple sign to hexadecimal place
```

The canonical sentence is:

```
Omicron chooses which sign exists  
and at which bounded low-gauge position it may be read.
```

7. Omnicron

7.1 Definition

Omicron is the larger four-plane circular runtime resolver.

It owns:

```
0x00..0x1F  
LOW CONTROL PLANE  
32 positions  
  
0x20..0x7F  
LOW AFFINE SPACE  
96 positions  
  
0x80..0x9F  
HIGH PROJECTIVE CONTROL PLANE  
32 positions  
  
0xA0..0xFF  
HIGH PROJECTIVE SPACE  
96 positions
```

Thus:

```
32 + 96 + 32 + 96 = 256
```

Omicron does not replace Omicron.

Omicron defines the bounded low gauge.

Omnicon resolves it against the full low/high byte field.

7.2 The Four-Plane Symmetry

The four planes form:

```
low control  
low affine  
high projective control  
high projective
```

The low and high halves share the same structural split:

```
32 control positions  
96 space positions
```

This creates:

```
low:  32 + 96 = 128  
high: 32 + 96 = 128
```

and:

```
128 + 128 = 256
```

The full byte field is therefore resolved as two complementary 128-position halves.

7.3 Omnicon as Runtime Resolver

Omnicon coordinates:

```
current plane  
circular position  
runtime cadence  
delta transition  
low/high correspondence
```

```
affine/projective correspondence  
candidate routing
```

It does not:

```
validate relation closure  
scribe incidence  
project azimuth  
record acceptance
```

It resolves where the system is.

It does not decide whether the relation is lawful.

7.4 Omnicron Canon

Omicron:

```
0x00..0x1F  
0x20..0x7F  
0x80..0x9F  
0xA0..0xFF
```

Role:

```
resolve the four circular runtime planes  
coordinate low/high correspondence  
coordinate affine/projective position  
schedule whole-system traversal
```

The canonical sentence is:

```
Omicron resolves the bounded Omicron gauge  
against its high projective complement  
and advances the whole-system circular runtime.
```

8. The Numerical Architecture

The witness families form a layered numerical architecture.

8.1 Relation Numbers

```
0x1E = 30  
0x3C = 60  
0x78 = 120
```

They form:

```
30  
→ doubled to 60  
→ doubled to 120
```

This is the relation fold ladder.

8.2 Incidence Numbers

```
0x18 = 24
```

This corresponds to:

```
4!  
24 gauge permutations  
24 Miquel flags
```

The hexadecimal gauge ladder is:

```
1  
16  
256  
4096  
65536
```

or:

```
0x0001  
0x0010  
0x0100
```

```
0x1000
0x10000
```

This is place escalation, not arithmetic doubling.

8.3 Projection Numbers

```
0xAA55
0x55AA
```

These are not scale magnitudes.

They are complementary orientation words.

Their importance comes from:

```
equal population
maximal alternation
exact complement
full XOR horizon
```

They belong to orientation rather than carry.

8.4 Sexagesimal Relation

The architecture contains two different decompositions of 60.

Incidence and observer surface:

```
24 + 36 = 60
```

where:

```
24
incidence flags

36
3 clocks × 3 bands × 4 offsets
```

Relation diagonals:

$$30 + 30 = 60$$

where:

$$\begin{array}{l} 30 \\ D^+ \text{ sum} \\ \\ 30 \\ D^- \text{ sum} \end{array}$$

The two decompositions are different:

$$\begin{array}{l} 24 + 36 \\ \text{incidence and observation} \\ \\ 30 + 30 \\ \text{relation and complementary relation} \end{array}$$

Both meet at the sexagesimal cycle.

8.5 The 7! Cycle

The full cycle is:

$$7! = 5040$$

It admits exact decompositions:

$$\begin{array}{l} 5040 = 84 \times 60 \\ \\ 5040 = 210 \times 24 \\ \\ 5040 = 168 \times 30 \\ \\ 5040 = 42 \times 120 \end{array}$$

These are not merely numerological decorations.

They show that the principal witness families fit exactly inside the finite scheduler:

```
60
sexagesimal phase

24
incidence block

30
relation diagonal

120
full relation fold
```

Every principal witness divides the 7! cycle without remainder.

That makes them usable as exact scheduling and validation periods.

9. Corrected Authority Flow

The corrected flow is:

1. Omnicron resolves runtime plane and cadence.
2. Omicron resolves bounded sign and place.
3. OMI-Lisp emits a lazy declaration.
4. OMI cites the addressed relation.
5. Tetragrammatron Relation Governor checks:
diagonal XOR closure,
0x1E relation sums,
0x3C centered fold,
0x78 shifted fold.
6. Metatron Incidence Scribe records:
FS/GS/RS/US position,
open/sealed polarity,
pairwise gauge plane,
0x18 incidence witness,
0x10000 gauge carry.

7. Receipt records:
relation verdict,
incidence record,
accepted or rejected state.
8. Gnomonic Projection Azimuth computes:
balanced orientation,
complementary phase,
azimuth,
tangent-plane projection.
9. IMO carries:
receipt,
bytes,
events,
files,
projected surfaces.

10. Witness Ownership

Tetragrammatron Relation Governor

0x1E
0x3C
0x78

Metatron Incidence Scribe

0x18

0x0001
0x0010
0x0100
0x1000
0x10000

Gnomonic Projection Azimuth

```
0xAA55  
0x55AA
```

Omicron

```
0x00..0x7F
```

Omnicon

```
0x00..0x1F  
0x20..0x7F  
0x80..0x9F  
0xA0..0xFF
```

11. Recommended Code Names

```
TetragrammatronRelationGovernor  
MetatronIncidenceScribe  
GnomonicProjectionAzimuth  
OmicronGauge  
OmniconResolver
```

Recommended prefixes:

```
tetra_relation_*  
metatron_incidence_*  
gnomonic_azimuth_*  
omicron_gauge_*  
omnicon_resolver_*
```

Example functions:

```
bool tetra_relation_validate_diagonal_xor(void);  
bool tetra_relation_validate_sum(void);  
bool tetra_relation_validate_centered_fold(void);
```



```
bool tetra_relation_validate_shifted_fold(void);

bool metatron_incidence_validate_flags(void);
void metatron_incidence_scribe_gauge(...);
void metatron_incidence_scribe_polarity(...);
void metatron_incidence_scribe_carry(...);

double gnomonic_azimuth_resolve(...);
GnomonicPoint gnomonic_azimuth_project(...);

OmicronPosition omicron_gauge_resolve(...);
OmicronState omnicron_resolver_advance(...);
```

12. Migration Rule

The renaming should be applied semantically, not through blind text replacement.

Use:

```
old Tetragrammatron relation logic
→ Tetragrammatron Relation Governor

old Tetragrammatron gauge/incidence logic
→ Metatron Incidence Scribe

old Metatron 0xAA55/0x55AA logic
→ Gnomonic Projection Azimuth
```

Do not rename every historical mention of Tetragrammatron to Metatron.

Only the incidence and gauge-polarity functions move.

Do not rename every historical mention of Metatron to Gnomonic Projection Azimuth.

Only the balanced alternating projection witness moves.

The migration must preserve provenance.

13. Final Canon

Tetragrammatron Relation Governor:

0x1E
0x3C
0x78

Role:

diagonal closure
centered fold
shifted/full-system fold

Metatron Incidence Scribe:

0x18

0x0001
0x0010
0x0100
0x1000
0x10000

Role:

gauge incidence
open/sealed polarity
hexadecimal-place stepping
gauge carry

Gnomonic Projection Azimuth:

0xAA55
0x55AA

Role:

alternating balanced witness
contrast
observer-facing orientation
projection azimuth

Omicron:

0x00..0x7F

Role:

bounded low gauge
four 32-position character sectors
hidden-to-readable tangent
F-column branch
character/place coupling

Omicron:

0x00..0x1F

0x20..0x7F

0x80..0x9F

0xA0..0xFF

Role:

four-plane circular runtime resolver
low/high correspondence
affine/projective correspondence
whole-system cadence

The final lock is:

Tetragrammatron closes relation.

Metatron scribes incidence.

Gnomonic Projection Azimuth orients projection.

Omicron defines the bounded gauge.

Omicron resolves the whole runtime.

Receipt records acceptance.

The names now follow the mathematics.